

Water Usage Effectiveness (WUE) Improvement Report

Jurong Hyperscale Campus · Data Centre · Singapore

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1. Background

Jurong Hyperscale Campus (JHC) rejects heat through eight cooling towers. In 2024 the campus used 240,000 m³ of potable water, a water usage effectiveness (WUE) of 1.9 litres per kWh of IT energy. The cooling towers ran at only three cycles of concentration, so a large share of the water was lost to blowdown. Customers increasingly ask for WUE disclosures in tenders.

2. Measures

- Cooling tower make-up switched from potable water to NEWater, Singapore's high-grade reclaimed water, which has very low dissolved solids.
- Cycles of concentration raised from 3 to 7, made possible by NEWater's low mineral content, side-stream filtration and automated inhibitor dosing.
- Blowdown water recovered for irrigation and toilet flushing.
- Condenser-water setpoints optimised to reduce evaporation at part load.
- Online monitoring of conductivity, ORP and biocide residual, with monthly Legionella sampling.

3. Results

KPI	2024	2025
Total water (m³/yr)	240,000	171,000
Potable water (m³/yr)	240,000	48,000
Cycles of concentration	3	7
WUE (L/kWh)	1.9	1.35

4. Lessons learned

- NEWater's low mineral content is what allows much higher cycles of concentration.
- Monitor corrosion closely when water chemistry changes; corrosion coupons flagged an early issue on galvanised surfaces.
- Energy and water trade off: a lower condenser-water temperature saves chiller energy but increases evaporation, so both must be optimised together.
- Legionella control must stay robust as cycles rise; ORP-based biocide dosing and monthly testing were introduced before cycles were increased.